MODULE 28 · DOMAIN 5 - NETWORKING AND CONTENT DELIVERY

VPC Fundamentals

CIDR sizing, subnets, route tables, NAT, IPv6, IPAM and Block Public Access.

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A subnet is not public because of a setting. It is public because its **route table** has a route to an internet gateway. Everything else follows from that.

VPC and subnet sizing

- A VPC CIDR is between **/16 and /28**. The **primary CIDR cannot be changed** after creation, though secondary CIDRs can be added.
- A subnet is between **/28 and /16**, must sit inside a VPC CIDR, and **belongs to exactly one Availability Zone**.
- **AWS reserves five addresses in every subnet**: the first four and the last one. A /28 has 16 addresses and **11 usable**.
- The reserved four are network, VPC router, **DNS at base plus two**, and one held for future use.

> ON THE EXAM
>
> Subnets cannot span Availability Zones. Multi-AZ always means multiple subnets, which is why every AWS service asks for a list of them.

Sizing in practice

- **Too small is unfixable.** You cannot enlarge a subnet, only create another, and the primary VPC CIDR is fixed.
- Plan for **three Availability Zones** from the start, even if you deploy into two today. Retrofitting a third is painful.
- Remember what consumes addresses: every ENI, every Fargate task, every Lambda function attached to the VPC, every endpoint.
- A subnet with no free addresses **fails scale-out silently**. Module 12's scenario 5 was exactly this.

> ON THE EXAM
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> Address exhaustion is the failure that appears months later, during an incident, when the Auto Scaling group cannot launch. Size generously.

Route tables

- Every route table has a **local route** covering the VPC CIDR. It cannot be deleted or modified, and it always wins for internal traffic.
- Every subnet is associated with **exactly one** route table. Without an explicit association it uses the **main** route table.
- **Longest prefix match wins.** A /24 route beats a /16 route for an address inside both.
- Where prefixes tie, **static routes beat propagated ones**. That matters with VPN and Direct Connect propagation.

> ON THE EXAM
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> The main route table is the default for anything you forget to associate. Leaving it permissive means a new subnet is accidentally public.

Internet gateways

- An **internet gateway** is attached to the VPC, is horizontally scaled and highly available, and costs nothing itself.
- A subnet becomes **public** when its route table sends 0.0.0.0/0 to the internet gateway.
- **The instance also needs a public IPv4 address**, either auto-assigned by the subnet or an Elastic IP. A route alone is not enough.
- The gateway performs **one-to-one NAT** between the private address and the public one. The instance never sees its public address.

> ON THE EXAM
>
> "My instance is in a public subnet and cannot reach the internet" is usually a missing public IP, not a missing route.

NAT gateways

- Lets instances in **private** subnets reach the internet outbound, while nothing on the internet can initiate a connection inward.
- It lives in a **public subnet**, needs an **Elastic IP**, and is a **zonal resource**. Module 14 showed why one per AZ matters.
- Scales from 5 Gbps to 100 Gbps automatically, and supports around **55,000 simultaneous connections per unique destination**.
- Charged **per hour and per gigabyte processed**, which makes it one of the largest line items in most VPC bills.

> ON THE EXAM
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> A NAT gateway in a private subnet cannot work, because it needs its own route to an internet gateway. That is a common misconfiguration.

The NAT gateway bill

- **Data processing is charged per gigabyte**, in addition to the hourly rate, and it applies to traffic to AWS services too.
- **VPC endpoints remove that traffic entirely.** A gateway endpoint for S3 or DynamoDB is free and bypasses the NAT gateway.
- **One NAT gateway per AZ** avoids paying cross-AZ data transfer on top of the processing charge.
- For a development environment, a single NAT gateway shared across zones is a legitimate cost-availability trade.

> ON THE EXAM
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> "Our NAT gateway bill is enormous and most of the traffic is S3" is a gateway endpoint. It is free, and it removes the traffic rather than reducing it.

NAT gateway or NAT instance?

NAT GATEWAY

Managed, and the default

- AWS manages availability and patching
- Scales automatically to 100 Gbps
- No security group; it is not an instance
- Charged hourly plus per GB

NAT INSTANCE

Legacy, occasionally justified

- An EC2 instance you patch and scale yourself
- Can be a bastion, or run custom software
- Needs source/destination check disabled
- Cheaper at very low volume

THE TIEBREAKER

A NAT instance needs source/destination check disabled or it silently drops forwarded traffic. That single setting is the classic exam detail.

IPv6 in a VPC

- An Amazon-provided VPC IPv6 block is a **/56**. Every subnet gets a **/64**, and that size is fixed.
- **IPv6 addresses are globally unique and publicly routable.** There is no private IPv6 equivalent of RFC 1918 by default.
- So outbound-only access uses an **egress-only internet gateway**, which is the IPv6 counterpart of a NAT gateway.
- It is **stateful and free**. There is no data processing charge, unlike a NAT gateway.

> ON THE EXAM
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> Because IPv6 addresses are public, adding IPv6 to a subnet can expose instances that were private over IPv4. Check your security groups.

The public IPv4 charge

- **Since February 2024, every public IPv4 address is charged hourly**, whether it is attached to anything or sitting idle.
- That covers EC2 public IPs, Elastic IPs, load balancers, NAT gateways, RDS and more. Roughly a few pounds per address per month.
- **Public IP Insights**, part of IPAM and free, shows every public IPv4 address in the account and what is using it.
- Reduce it by **releasing idle Elastic IPs**, moving to IPv6 where possible, and not auto-assigning public IPs to instances that do not need them.

> ON THE EXAM
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> Idle Elastic IPs used to be the only ones charged. Since 2024 every public IPv4 address costs money, attached or not.

VPC IP Address Manager

- **IPAM** plans, tracks and monitors IP address usage across accounts and Regions, replacing the spreadsheet.
- **Pools** are hierarchical, so a top-level range is subdivided per Region, per environment and per account, with automatic allocation.
- The **free tier** covers basic monitoring and Public IP Insights. The **advanced tier** adds cross-account, cross-Region and BYOIP management.
- **Amazon-provided contiguous IPv4 blocks** give sequential Elastic IPs, so a partner can allow-list one range rather than thirty addresses.

> ON THE EXAM
>
> "We cannot tell which CIDR ranges are in use across 60 accounts" is IPAM. It is in the exam's in-scope list precisely because that problem is universal.

VPC Block Public Access

- An **account and Region level** control that blocks internet traffic to and from VPC resources, regardless of route tables and security groups.
- Two modes: **Block Ingress**, where inbound from the internet is blocked but NAT and egress-only gateways still work, and **Block Bidirectional**.
- **Exclusions** allow specific VPCs or subnets to remain reachable, so it is not all-or-nothing.
- Deployable **organisation-wide as a declarative policy**, as module 25 covered, so new accounts inherit it automatically.

> ON THE EXAM
>
> This is a switch above the route table. It overrides an accidentally public subnet without anyone having to find and fix the route.

VPC Encryption Controls

- Enforces that traffic within and between your VPCs is **encrypted in transit**, as a network-level control rather than an application setting.
- Also deployable through a **declarative policy**, so the requirement applies organisation-wide.
- It was **free until 1 March 2026**, and is now charged **per non-empty VPC per hour**, at a rate that varies by Region.
- Useful where a compliance regime demands encryption in transit everywhere, including between instances inside one VPC.

> ON THE EXAM
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> Compliance frameworks increasingly require encryption in transit internally, not just at the edge. This enforces it without touching applications.

The remaining pieces

- An **elastic network interface** is the thing that actually holds addresses and security groups. Instances have one or more.
- **DHCP option sets** control the DNS servers, domain name and NTP servers handed to instances. Useful with on-premises DNS.
- **enableDnsSupport** and **enableDnsHostnames** on the VPC decide whether the Amazon resolver works and whether instances get DNS names.
- The **default VPC** has public subnets in every AZ and auto-assigns public IPs. Convenient, and not what production should use.

> ON THE EXAM
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> Interface endpoints and Route 53 private hosted zones both depend on enableDnsSupport. If DNS resolution fails inside a VPC, check those two flags first.

> An EC2 instance in a subnet whose route table sends 0.0.0.0/0 to an internet gateway cannot reach the internet. The security group allows all outbound traffic, and the network ACL is the default allow-all.
>
> The instance has only a private IPv4 address.

What is the cause?

A The instance has no public IPv4 address, so the internet gateway has nothing to translate to

B The subnet needs a NAT gateway as well as an internet gateway

C The route table needs a route to the NAT gateway

D The internet gateway must be attached to the subnet

◆ PAUSE AND DECIDE

What is the cause?

A No public IPv4 address

An internet gateway performs one-to-one NAT between a private and a public address. With no public address there is nothing to translate.

B Needs a NAT gateway too

A NAT gateway is for instances in private subnets. This subnet already routes to an internet gateway, which is the public pattern.

C Route to a NAT gateway

Would give the subnet two default routes to different targets, which is not valid. It also does not address the missing address.

D Attach the IGW to the subnet

Internet gateways attach to a VPC, not a subnet. Subnets connect to one through a route table.

THE SIGNAL

Route plus public address, both required. The scenario eliminates security groups and NACLs deliberately, leaving only the addressing.

> A company's NAT gateway charges have grown to a significant monthly figure. Analysis shows that about 80 percent of the processed traffic is private instances reading from and writing to S3 buckets, in the same Region.
>
> The team wants to reduce this without moving the instances to public subnets.

What should the team implement?

A. A gateway VPC endpoint for S3, with routes added to the private subnets' route tables

B. A second NAT gateway to spread the load

C. Move the buckets to another Region

D. Replace the NAT gateway with a NAT instance

◆ PAUSE AND DECIDE

What should the team implement?

A Gateway endpoint for S3

Gateway endpoints are free and route S3 traffic directly, removing 80 percent of the volume from the NAT gateway rather than reducing it.

B A second NAT gateway

Doubles the hourly charge and processes the same total volume. It increases cost, not reduces it.

C Move the buckets

Cross-Region traffic is more expensive, not less, and it adds latency. It makes the problem worse in both dimensions.

D NAT instance

Trades a managed service for one you patch and scale, and still carries all the traffic. The volume is the cost driver, not the appliance.

THE SIGNAL

Gateway endpoints for S3 and DynamoDB are free. That is why this is the single highest-value networking cost answer on the exam.

> An Auto Scaling group has stopped launching instances during peak periods. The scaling activity history reports 'There are not enough free addresses in subnet subnet-0a1b2c'. The subnet is a /27 and also hosts an interface endpoint and several Fargate tasks.
>
> The VPC's primary CIDR is a /24 and is fully allocated to existing subnets.

What is the most appropriate fix?

A Enlarge the subnet from /27 to /25

B Add a secondary CIDR block to the VPC, create larger subnets in it, and move the Auto Scaling group to them

C Change the VPC's primary CIDR to a /16

D Release unused Elastic IP addresses in the account

◆ PAUSE AND DECIDE

What is the most appropriate fix?

A. Enlarge the subnet

Subnets cannot be resized. The only way to get a larger subnet is to create a new one, and the existing /24 has no space.

B. Secondary CIDR, new subnets, migrate

The primary CIDR is immutable and full, so a secondary block is the only route to more address space. Then migrate.

C. Change the primary CIDR

The primary CIDR cannot be changed after the VPC is created. This is not an option that exists.

D. Release Elastic IPs

Elastic IPs are public addresses. The shortage here is private addresses inside a subnet, which is unrelated.

THE SIGNAL

Two immutable facts do the work: subnets cannot be resized, and a primary VPC CIDR cannot be changed. Both A and C describe operations that do not exist.

> A security team must guarantee that no resource in any VPC across the organisation can be reached from the internet. A small number of documented exceptions are allowed. Route tables and security groups are managed by many different application teams.
>
> New accounts are created weekly and must be covered automatically.

What should the team implement?

A. VPC Block Public Access in Block Ingress mode with exclusions, deployed as a declarative policy

B. An SCP denying ec2:CreateRoute for internet gateway destinations

C. A Config rule detecting route tables with an internet gateway route

D. Delete all internet gateways in all accounts

◆ PAUSE AND DECIDE

What should the team implement?

A Block Public Access via declarative policy

Overrides route tables and security groups directly, supports documented exclusions, and a declarative policy covers accounts created later.

B SCP denying CreateRoute

Stops new routes being created and does nothing about routes that already exist. It also blocks legitimate exception cases entirely.

C Config rule

Detective. It reports a publicly reachable resource after it exists, and the requirement is that none can be reached.

D Delete all internet gateways

Breaks NAT gateways and every legitimate outbound path, and the exceptions could not be accommodated at all.

THE SIGNAL

Block Ingress is the mode that matters here: inbound is blocked while NAT and egress-only gateways keep working, so outbound access survives.

> A company with 70 accounts cannot determine which private CIDR ranges are already in use. Two teams recently created overlapping VPCs, which blocked a Transit Gateway attachment. Address planning is currently a shared spreadsheet.
>
> They also want to identify and release unused public IPv4 addresses, which now carry an hourly charge.

What should the company adopt?

A. VPC IPAM with hierarchical pools for allocation, plus Public IP Insights for the IPv4 clean-up

B. A Config rule recording every VPC CIDR into an aggregator

C. Reachability Analyzer across all accounts

D. A tag policy requiring every VPC to be tagged with its CIDR

◆ PAUSE AND DECIDE

What should the company adopt?

A IPAM pools plus Public IP Insights

Pools allocate non-overlapping ranges automatically, which prevents the collision rather than reporting it, and Public IP Insights is free and covers the second requirement.

B Config rule into an aggregator

Gives visibility of what exists. It does not allocate ranges, so two teams could still pick overlapping CIDRs on the same day.

C Reachability Analyzer

Tests whether a path exists between two resources. It is a troubleshooting tool, not an address planning one.

D Tag policy

A spreadsheet in tag form. It records intent and prevents nothing, and it goes stale exactly as the spreadsheet did.

THE SIGNAL

Preventing the overlap beats detecting it. IPAM allocates from a pool, so two teams cannot be handed the same range.

The five things to carry forward

- **A subnet is public because of its route table.** There is no setting, and an instance also needs a public IP.
- **Five reserved addresses per subnet.** A /28 gives 11 usable, and subnets cannot be resized.
- **Gateway endpoints for S3 and DynamoDB are free** and remove that traffic from the NAT gateway entirely.
- **Every public IPv4 address is charged hourly since February 2024**, attached or idle.
- **Block Public Access overrides route tables**, and IPAM prevents CIDR overlap rather than reporting it.

> ON THE EXAM
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> Next: security groups, NACLs, and a systematic ladder for troubleshooting connectivity.